Cryptography Notes

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Contents

1 Divisibility

Proposition 1.1. *Let $a, b, c \in \mathbb{Z}$. The following properties hold:*

1. *If $a \mid 1$, then $a = \pm 1$.*
2. *If $a \mid b$ and $b \mid a$, and $a \neq 0, b \neq 0$, then $a = \pm b$.*
3. *If $a \mid b$ and $b \mid c$, then $a \mid c$ (Transitivity).*

Proof.

1. If $a \mid 1$: By definition, $\exists c \in \mathbb{Z} : 1 = ac$. Since $a, c \in \mathbb{Z}$, the only possible integer solutions are $(a = 1, c = 1)$ or $(a = -1, c = -1)$. Therefore, $a = \pm 1$.

2. If $a \mid b$ and $b \mid a$: By definition, $\exists c_1, c_2 \in \mathbb{Z} : b = ac_1$ and $a = bc_2$. Substituting the first equation into the second gives $a = (ac_1)c_2 = a(c_1c_2)$. Since $a \neq 0$, we can divide by a to get $1 = c_1c_2$. As $c_1, c_2 \in \mathbb{Z}$, the only solutions are $(c_1 = 1, c_2 = 1)$ or $(c_1 = -1, c_2 = -1)$.

   - If $c_1 = 1$, then $b = a \cdot 1 = a$.
   - If $c_1 = -1$, then $b = a \cdot (-1) = -a$.

   Thus, $a = \pm b$.

3. If $a \mid b$ and $b \mid c$: By definition, $\exists c_1, c_2 \in \mathbb{Z} : b = ac_1$ and $c = bc_2$. Substituting the first equation into the second gives $c = (ac_1)c_2 = a(c_1c_2)$. Since $c_1c_2 \in \mathbb{Z}$, this satisfies the definition of divisibility, so $a \mid c$. □

Proposition 1.2 (Euclidean Division). *Let $a \in \mathbb{N}$ and $b \in \mathbb{N}$ with $b > 0$. Then $\exists! q, r \in \mathbb{Z} : a = bq + r$ with $0 \leq r < b$.*

Proposition 1.3. *Let $m \in \mathbb{N}$, with $m < 2^B$. The repeated division-by-2 process:*

$$\begin{aligned} m &= 2q_0 + m_0 \\ q_0 &= 2q_1 + m_1 \\ q_1 &= 2q_2 + m_2 \\ &\dots \end{aligned}$$

terminates with $q_{B-1} = 0$, and $m = m_0 + 2m_1 + \cdots + 2^{B-1}m_{B-1}$.

Proof. The proof consists of two parts: demonstrating the polynomial form and proving that the process terminates.

Polynomial Form

We can show the relationship by algebraic substitution.

$$m = 2q_0 + m_0$$

Substitute $q_0 = 2q_1 + m_1$:

$$m = 2(2q_1 + m_1) + m_0 = 2^2 q_1 + 2m_1 + m_0$$

Substitute $q_1 = 2q_2 + m_2$:

$$m = 2^2(2q_2 + m_2) + 2m_1 + m_0 = 2^3 q_2 + 2^2 m_2 + 2m_1 + m_0$$

Continuing this process until the final step where $q_{B-2} = 2q_{B-1} + m_{B-1}$, we arrive at the final form: $m = m_0 + 2m_1 + 2^2 m_2 + \cdots + 2^{B-1} m_{B-1}$.

Termination

The process is guaranteed to terminate because the sequence of quotients is strictly decreasing and bounded.

- Since $m < 2^B$, the first division $m = 2q_0 + m_0$ implies $2q_0 \le m < 2^B$, so $q_0 < 2^{B-1}$.
- Similarly, $q_0 < 2^{B-1}$ implies $2q_1 \le q_0 < 2^{B-1}$, so $q_1 < 2^{B-2}$.

This pattern continues: $q_i < 2^{B-1-i}$. For the final quotient q_{B-1}, we have $q_{B-1} < 2^{B-1-(B-1)} = 2^0 = 1$. Since $q_{B-1} \in \mathbb{N} \cup \{0\}$, the only possibility is $q_{B-1} = 0$. The process terminates when the quotient becomes zero. $\square$

Remark. *If m has B bits in its binary representation, i.e., $m = (m_{B-1} \dots m_0)_2$, then $m < 2^B$.*

Proof. The largest possible value for a B-bit number occurs when all bits are 1. This value can be calculated by summing the geometric series:

$$m \le \sum_{i=0}^{B-1} 2^i = \frac{2^B - 1}{2 - 1} = 2^B - 1$$

Therefore, $m < 2^B$. $\square$

Corollary 1.1. *A natural number $m \in \mathbb{N}$ has a binary representation with B bits (where the most significant bit $m_{B-1} = 1$) $\iff 2^{B-1} \le m < 2^B$.*

2 The Greatest Common Divisor (GCD)

Definition 2.1. *Let $a, b \in \mathbb{Z}$, such that $a \neq 0$ or $b \neq 0$. The set of their common divisors has a largest element $d \in \mathbb{N}$, called the greatest common divisor of a and b. We write $d = \gcd(a, b)$.*

Remark. *To find $\gcd(18, 12)$, we can list the divisors of each number:*

- *Divisors of 12:* $\{1, 2, 3, 4, 6, 12\}$
- *Divisors of 18:* $\{1, 2, 3, 6, 9, 18\}$

The set of common divisors is $\{1, 2, 3, 6\}$. The largest element in this set is 6, so $\gcd(18, 12) = 6$.

While listing all divisors is feasible for small numbers, it is impractical for large integers. A far more powerful and efficient method is the Euclidean Algorithm.

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3 The Euclidean Algorithm

Theorem 3.1 (Euclidean Algorithm). *Let $a, b \in \mathbb{N} : a > 0, b > 0$ and $a \geq b$. The following algorithm computes $\gcd(a, b)$ in a finite number of steps:*

1. *Let $r_0 := a$ and $r_1 := b$.*
2. *Set $i = 1$.*
3. *Divide r_{i-1} by r_i to obtain a quotient q_i and remainder r_{i+1}: $r_{i-1} = r_i q_i + r_{i+1}$.*
4. *If $r_{i+1} = 0$, then $r_i = \gcd(a, b)$. The algorithm terminates.*
5. *Otherwise, increment $i \to i + 1$ and return to Step 3.*

Proof. Let the sequence of remainders be $r_0, r_1, \ldots, r_n, r_{n+1}$ where $d = r_n$ is the last non-zero remainder and $r_{n+1} = 0$. We must demonstrate two facts: that d is a common divisor of a and b, and that it is the greatest common divisor.

a. $d \mid a$ **and** $d \mid b$

We proceed by working backward from the end of the algorithm.

- The final division step is $r_{n-1} = r_n q_n + r_{n+1}$. Since $r_{n+1} = 0$ and $d = r_n$, this equation becomes $r_{n-1} = dq_n$. This demonstrates $d \mid r_{n-1}$.
- The preceding step is $r_{n-2} = r_{n-1} q_{n-1} + r_n$. Since $d \mid r_n$ and we have shown $d \mid r_{n-1}$, it follows that d must divide their linear combination, $d \mid r_{n-2}$.

Continuing this backward induction process, we establish that d divides every preceding remainder. This concludes that $d \mid r_1$ (where $r_1 = b$) and $d \mid r_0$ (where $r_0 = a$). Thus, d is a common divisor.

b. d **is the Greatest Common Divisor**

Let $c \in \mathbb{Z}$ be any common divisor of a and b. We work forward down the chain of remainders.

- Since $c \mid a$ (r_0) and $c \mid b$ (r_1), and $r_2 = r_0 - r_1 q_1$, it must be that $c \mid r_2$.
- Since $c \mid r_1$ and $c \mid r_2$, and $r_3 = r_1 - r_2 q_2$, it must be that $c \mid r_3$.

Continuing this process, we eventually find that c must divide the last non-zero remainder, r_n, so $c \mid d$. Since $c \mid d$, it is a consequence of divisibility that $c \leq d$. This holds $\forall c$ common divisor, proving that d is the greatest common divisor. $\square$

Proposition 3.1 (Extended Euclidean Algorithm). *Let $a, b \in \mathbb{N}$ such that $a \neq 0$ or $b \neq 0$. If $d = \gcd(a, b)$, then $\exists u, v \in \mathbb{Z} : d = au + bv$.*

Proposition 3.2 (Solvability Condition). *The linear Diophantine equation $au + bv = c$ has integer solutions $u, v \iff \gcd(a, b) \mid c$.*

Proof. (⇒) Assume integer solutions u, v exist. Let $d = \gcd(a, b)$. By definition, $d \mid a$ and $d \mid b$. $\implies d \mid (au + bv)$, $\implies d \mid c$ because $au + bv = c$.

(⇐) Assume $d = \gcd(a, b)$ and $d \mid c$. Then $\exists k \in \mathbb{Z} : c = kd$. From the Extended Euclidean Algorithm, we know $\exists u_0, v_0 \in \mathbb{Z} : au_0 + bv_0 = d$. Multiplying by k, we get $a(ku_0) + b(kv_0) = kd = c$. Thus, a solution exists with $u = ku_0$ and $v = kv_0$. □

Proposition 3.3 (Form of All Solutions). *If (u_p, v_p) is a particular solution to $au + bv = c$ and $d = \gcd(a, b)$, let $a' = a/d$ and $b' = b/d$. Then $\forall (u, v)$ integer solutions, $u = u_p + kb'$ and $v = v_p - ka'$ for any $k \in \mathbb{Z}$.*

Proof. Assume (u', v') is an arbitrary integer solution, so $au' + bv' = c$. We also have $au_p + bv_p = c$. Subtracting the two equations gives $a(u' - u_p) + b(v' - v_p) = 0$, so $a(u' - u_p) = b(v_p - v')$. Dividing by d: $a'(u' - u_p) = b'(v_p - v')$. Since $\gcd(a', b') = 1$, and $a' \mid b'(v_p - v')$, it must be that $a' \mid (v_p - v')$. $\implies \exists k \in \mathbb{Z} : v_p - v' = ka'$, which gives $v' = v_p - ka'$. Substituting this back yields $a'(u' - u_p) = b'(ka')$, which simplifies to $u' - u_p = kb'$, or $u' = u_p + kb'$. This demonstrates that any other solution must be of the specified form. □

Proposition 3.4 (Characterisation of Coprime Numbers). *Integers $a, b \in \mathbb{Z}$ are relatively prime $\iff \exists u, v \in \mathbb{Z} : au + bv = 1$.*

Proof. ($\Rightarrow$) This is a direct consequence of the Extended Euclidean Algorithm. If $\gcd(a, b) = 1$, then the algorithm guarantees $\exists u, v \in \mathbb{Z} : au + bv = 1$.

($\Leftarrow$) Conversely, assume $\exists u, v \in \mathbb{Z} : au + bv = 1$. Let $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, it must divide their linear combination $au + bv$. Therefore, $d \mid 1$. The only $d \in \mathbb{N}$ so that $d \mid 1$ is $d = 1$. □

Corollary 3.1. *Let p be a prime number and $m \in \mathbb{Z}$. If $p \nmid m$, then $\gcd(p, m) = 1$.*

Proof. The positive divisors of a prime number p are only $\{1, p\}$. If $p \nmid m$, then p is not a common divisor. Therefore, the only common divisor of p and m is 1, which means $\gcd(p, m) = 1$. □

Proposition 3.5 (Prime Divisor Property). *Let p be a prime and $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. Assume $p \mid ab$. If $p \mid a$, the proposition holds. If $p \nmid a$, then by the preceding corollary, $\gcd(p, a) = 1$. By the characterization of coprime numbers, $\exists u, v \in \mathbb{Z} : pu + av = 1$. Multiplying this equation by b gives $pub + abv = b$.

Since $p \mid pub$ and we are given $p \mid ab$, which implies $p \mid abv$. Since p divides both terms on the left side, it must divide their sum. Therefore, $p \mid (pub + abv) \to p \mid b$. □

Theorem 3.2 (Fundamental Theorem of Arithmetic). *Let $a \in \mathbb{N}$ so that $a \geq 2$. Then a can be factored as a product of primes, $a = p_1^{e_1} p_2^{e_2} \cdots p_m^{e_m}$, where p_i are distinct primes, and $\forall i \in \{1, \ldots, m\}, e_i \geq 1$. The factorization into primes is unique modulo rearrangements of the order of the factors.*

Remark. *The exponent $e_i = mult_{p_i}(a)$ is the multiplicity of p_i in a. A generalized form of the prime factorization is given by:*

$$a = \prod_{p \text{ prime}} p^{mult_p(a)}$$

where $mult_p(a) := 0$ if p is not a prime factor of a.

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4 Classes Modulo m

Proposition 4.1. *$\forall a \in \mathbb{Z}$ (the dividend) and $\forall m \in \mathbb{N}$ so that $m > 1$ (the modulus), $\exists! q, r \in \mathbb{Z}$: $a = mq + r$, with $0 \leq r < m$.*

Theorem 4.1 (Existence of a Unique Remainder)**.** *$\forall a \in \mathbb{Z}$ and $\forall m \in \mathbb{N} : m > 1$, $\exists! r \in \{0, 1, \ldots, m-1\} : a \equiv r \pmod{m}$.*

Proof. By Proposition 4.1 (Euclidean Division), we have $\exists! q, r \in \mathbb{Z} : a = mq + r$, where $0 \leq r < m$. Rearranging this equation gives $a - r = mq$. By the formal definition of divisibility, $m \mid (a - r)$. By the formal definition of congruence, this directly implies $a \equiv r \pmod{m}$. The uniqueness of r is guaranteed by the Euclidean Division Theorem. □

Theorem 4.2 (Congruence is an Equivalence Relation)**.** *The relation of congruence modulo m (denoted by $\equiv \pmod{m}$) is an equivalence relation on the set of integers $\mathbb{Z}$.*

Proof. We must demonstrate that congruence modulo m satisfies reflexivity, symmetry, and transitivity.

- **Reflexivity:** $\forall a \in \mathbb{Z}$, $a - a = 0$. Since $m \mid 0$, $a \equiv a \pmod{m}$. The relation is reflexive.
- **Symmetry:** Assume $a \equiv b \pmod{m}$. $\implies m \mid (a - b)$, so $\exists k \in \mathbb{Z} : a - b = km$. Multiplying by -1 yields $b - a = (-k)m$. Since $-k \in \mathbb{Z}$, $\implies m \mid (b - a)$, so $b \equiv a \pmod{m}$. The relation is symmetric.
- **Transitivity:** Assume $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. $\implies \exists k_1, k_2 \in \mathbb{Z} : a - b = k_1 m$ and $b - c = k_2 m$. Adding the equations:

$$(a - b) + (b - c) = k_1 m + k_2 m$$

$$a - c = (k_1 + k_2)m$$

  Since $k_1 + k_2 \in \mathbb{Z}$, $\implies m \mid (a - c)$, so $a \equiv c \pmod{m}$. The relation is transitive. □

Proposition 4.2 (Equivalence Classes Form a Partition)**.** *Let $\sim$ be an equivalence relation on a set X. The set of equivalence classes, $X/\sim$, forms a partition of X.*

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5 Invertibility Modulo m

Proposition 5.1 (Compatibility)**.** *Let* $a_1, a_2, b_1, b_2 \in \mathbb{Z}$*. If* $a_1 \equiv a_2 \pmod{m}$ *and* $b_1 \equiv b_2 \pmod{m}$*, then:*

1. $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$
2. $a_1 b_1 \equiv a_2 b_2 \pmod{m}$

Proof.

- **Addition:** Given $a_2 \equiv a_1 \pmod{m}$ and $b_2 \equiv b_1 \pmod{m}$, $\implies \exists k_1, k_2 \in \mathbb{Z}$: $a_2 - a_1 = k_1 m$ and $b_2 - b_1 = k_2 m$. Adding the equations yields $(a_2 - a_1) + (b_2 - b_1) = (k_1 + k_2)m$. Rearranging the terms: $(a_2 + b_2) - (a_1 + b_1) = (k_1 + k_2)m$. Since $k_1 + k_2 \in \mathbb{Z}$, this shows $m \mid ((a_2 + b_2) - (a_1 + b_1))$, so $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$.
- **Multiplication:** We express $a_2 = a_1 + k_1 m$ and $b_2 = b_1 + k_2 m$. Multiplying them gives:

$$a_2 b_2 = (a_1 + k_1 m)(b_1 + k_2 m) = a_1 b_1 + a_1 k_2 m + b_1 k_1 m + k_1 k_2 m^2$$

Factoring out m: $a_2 b_2 - a_1 b_1 = m(a_1 k_2 + b_1 k_1 + k_1 k_2 m)$. Let $K = a_1 k_2 + b_1 k_1 + k_1 k_2 m$. Since $K \in \mathbb{Z}$, $m \mid (a_2 b_2 - a_1 b_1)$, so $a_1 b_1 \equiv a_2 b_2 \pmod{m}$. □

Proposition 5.2. *The set of integers modulo* m *under addition,* $(\mathbb{Z}/m\mathbb{Z}, +)$*, forms a **commutative group**.*

Proposition 5.3. *When considering both addition and multiplication,* $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$ *forms a **commutative ring**.*

Theorem 5.1. *If an inverse exists, it is unique.*

Proof. Assume an element $[a]$ has two inverses, $[a']$ and $[a'']$. By definition, $[a] \cdot [a'] = [1]$ and $[a] \cdot [a''] = [1]$. Since multiplication is associative and $[1]$ is the identity element, we can write:

$$[a'] = [a'] \cdot [1] = [a'] \cdot ([a] \cdot [a''])$$

$$[a'] = ([a'] \cdot [a]) \cdot [a''] = [1] \cdot [a''] = [a'']$$

Therefore, $[a'] = [a'']$, $\implies$ the inverse is unique. □

Theorem 5.2 (Characterization of Invertibility)**.** *An element* $[a] \in \mathbb{Z}/m\mathbb{Z}$ *is invertible* $\iff$ $\gcd(a, m) = 1$.

Proof. An element $[a]$ is invertible $\iff$ $\exists u \in \mathbb{Z} : au \equiv 1 \pmod{m}$. This congruence is equivalent to the linear Diophantine equation $au - 1 = -mv$ for some $v \in \mathbb{Z}$, which can be rewritten as $au + mv = 1$. By Bézout's Identity, this equation has an integer solution $\iff$ $\gcd(a, m) \mid 1$. Since $\gcd(a, m) \in \mathbb{N}$, this condition is satisfied $\iff$ $\gcd(a, m) = 1$. □

Proposition 5.4. *The set of units under multiplication, $(\mathbb{Z}/m\mathbb{Z})^*$, forms a* ***commutative group*** *(the group of units modulo m).*

Proposition 5.5. *$\forall p$ prime number, the commutative ring $(\mathbb{Z}/p\mathbb{Z}, +, \cdot)$ is a* ***field****, commonly denoted $\mathbb{F}_p$.*

Corollary 5.1. *Let $p \in \mathbb{N}$ be a prime so that $p > 2$. The equation $x^2 \equiv 1 \pmod p$ in $\mathbb{F}_p$ has exactly two distinct solutions: $x \equiv 1 \pmod p$ and $x \equiv p-1 \pmod p$ (i.e., $x \equiv -1 \pmod p$).*

Proof. The equation $x^2 \equiv 1 \pmod p$ is equivalent to $p \mid (x^2 - 1)$, or $p \mid (x-1)(x+1)$. Since p is prime, by the Prime Divisor Property:

- Either $p \mid (x - 1)$, which implies $x \equiv 1 \pmod p$.
- Or $p \mid (x + 1)$, which implies $x \equiv -1 \equiv p - 1 \pmod p$.

These two solutions are distinct because $1 \not\equiv -1 \pmod p$ for $p > 2$. $\square$

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6 Solving Linear Congruences

Proposition 6.1. *The linear congruence* $ax \equiv b \pmod{m}$ *has at least one integer solution for* $x \in \mathbb{Z} \iff d = \gcd(a, m) \mid b$.

Proof. The congruence $ax \equiv b \pmod{m}$ is equivalent to the linear Diophantine equation $ax + mv = b$ for some $v \in \mathbb{Z}$. This equation is solvable for $x, v \in \mathbb{Z} \iff$ the greatest common divisor of the coefficients, $d = \gcd(a, m)$, divides the constant term b. This follows directly from the Solvability Condition for Diophantine Equations (Bézout's identity). $\square$

Proposition 6.2. *If* $d = \gcd(a, m) \mid b$*, the congruence* $ax \equiv b \pmod{m}$ *has exactly* d *distinct solutions modulo* m*.*

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7 Chinese Remainder Theorem (CRT)

Theorem 7.1 (Chinese Remainder Theorem)**.** *Let $m_1, m_2, \ldots, m_k \in \mathbb{N}$ be pairwise relatively prime integers (i.e., $\forall i \neq j, \gcd(m_i, m_j) = 1$). Let $a_1, a_2, \ldots, a_k \in \mathbb{Z}$ be arbitrary integers. The system of congruences:*

$$x \equiv a_i \pmod{m_i} \quad \forall i \in \{1, \ldots, k\}$$

has a solution, and the complete set of solutions is congruent modulo the product $M = m_1 \cdot m_2 \cdot \cdots \cdot m_k$.

Proof of the Chinese Remainder Theorem. The proof is divided into two parts: Existence (constructive) and Uniqueness (based on coprimality).

7.1 Proof of Existence

We define the solution constructively as $x = \sum_{j=1}^{k} a_j v_j N_j$. For this construction, let $M = \prod_{i=1}^{k} m_i$. We define $N_j = M/m_j$. Since the m_i are pairwise coprime, $\gcd(N_j, m_j) = 1$. $\implies \exists v_j \in \mathbb{Z} : v_j N_j \equiv 1 \pmod{m_j}$ (the multiplicative inverse of N_j modulo m_j). We must show that $x \equiv a_i \pmod{m_i}$ for an arbitrary index $i \in \{1, \ldots, k\}$.

The core argument is that $\forall j \neq i$, the term N_j contains m_i as a factor (since m_i is a factor of M but not m_j).

$$\forall j \neq i, \quad m_i \mid N_j \quad \implies a_j v_j N_j \equiv 0 \pmod{m_i}$$

Thus, when x is taken modulo m_i, all terms in the sum vanish except the i-th term ($j = i$):

$$x \equiv a_i v_i N_i + \sum_{j \neq i} (0) \pmod{m_i}$$

$$x \equiv a_i v_i N_i \pmod{m_i}$$

Since $v_i N_i \equiv 1 \pmod{m_i}$ by construction, the proof concludes:

$$x \equiv a_i(1) \equiv a_i \pmod{m_i}$$

This confirms the existence of a solution x that satisfies the system.

7.2 Proof of Uniqueness

1. **Assume Two Solutions:** Let x and x' both solve the system.
2. **Relationship:** $\forall i \in \{1, \ldots, k\}, x \equiv a_i \pmod{m_i}$ and $x' \equiv a_i \pmod{m_i}$. $\implies x - x' \equiv 0 \pmod{m_i}$. This means that $\forall i, m_i \mid (x - x')$.
3. **Use Co-prime Condition:** Since the moduli $m_1, \ldots, m_k$ are pairwise co-prime and each divides the difference $(x - x')$, their product $M = \prod_{i=1}^{k} m_i$ must also divide the difference.
4. **Conclusion:** $M \mid (x - x')$ means $x \equiv x' \pmod{M}$.

This confirms that the solution is unique modulo M. □

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8 Euler's ϕ Function

Theorem 8.1. *The Totient of a Prime Number: If $m \in \mathbb{N}$ so that m is prime, then $\varphi(m) = m-1$.*

Theorem 8.2. *A Primality Test via the Totient Function: $\forall m \in \mathbb{N} : m \geq 2$, $\varphi(m) = m-1 \iff m$ is a prime number.*

Proof. ($\Rightarrow$) **("Only if" direction)**: We prove the contrapositive. Assume m is composite (not prime). $\implies \exists p \in \mathbb{N} : 1 < p < m$ and $p \mid m$. Since $p \mid m$, $\gcd(p, m) = p > 1$. Thus, p is not coprime to m. Since $p \in \{1, \dots, m-1\}$ is excluded from the count, the total count $\varphi(m)$ must be strictly less than $m-1$. More precisely, $\varphi(m) \leq m-2$. This shows that the equality $\varphi(m) = m-1$ holds $\implies m$ is prime.

($\Leftarrow$) **("If" direction)**: Assume m is prime. Then $\forall a \in \{1, \dots, m-1\}$, $\gcd(a, m) = 1$. Since there are $m-1$ such integers, $\varphi(m) = m-1$. ☐

Proposition 8.1. *$\forall p$ prime and $\forall \alpha \in \mathbb{N} : \alpha \geq 1$:*

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$$

Proof. The totient function $\varphi(p^\alpha)$ counts the number of integers $a \in \{1, \dots, p^\alpha\}$ so that $\gcd(a, p^\alpha) = 1$. The integers that are **not** coprime to p^α are those a for which $p \mid a$. We count these integers (the complement). These non-coprime integers are the multiples of p in the range $[1, p^\alpha]$, i.e., of the form $k \cdot p$ where $k \in \mathbb{N}$. The condition $1 \leq k \cdot p \leq p^\alpha$ implies $1 \leq k \leq p^{\alpha-1}$. Thus, there are exactly $p^{\alpha-1}$ multiples of p in the specified range. Subtracting this count from the total number of integers, p^α, yields:

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1)$$

☐

Theorem 8.3. *Multiplicativity of Euler's Totient Function: If $m, n \in \mathbb{N}$ are coprime integers ($\gcd(m, n) = 1$), then $\varphi(mn) = \varphi(m)\varphi(n)$.*